Supporting Material: Artifact-Level Detail for Rank Instability and Representation Repair in Automatic Modulation Classification under Structured Interference

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Abstract

This is optional supporting material. No result required to support the manuscript's main claims appears only in this document. It provides complete cell construction, aggregation definitions, per-seed/SNR/class results, architecture details, frozen-control analyses, convergence histories, prospective-freeze provenance, and artifact paths for results summarized in the manuscript. The four sequential severe-held stages—the transfer check, the sidecar repair, the inference-time attribution controls, and the retrained zeroed-I/Q ablation—are stated in the manuscript and expanded here at artifact level. The document additionally reports the evaluation-cell construction, capacity-control sweep, comparator-fidelity curves, envelope audits, severe-held construction, attribution controls, and seed-level robustness.

I. Scope and How to Read This Document

The study asks why compact spectral and high-capacity I/Q AMC representations exhibit condition-dependent rank instability, how far a campaign-internal empirical envelope transports, and whether a lightweight received-I/Q sidecar can repair the degradation exposed by independent severe-domain validation. All evidence is from source-disjoint 3GPP TR 38.901 TDL-profile simulation.

Because this document is standalone, the identifiers used throughout are restated here.

- **A0** is the direct spectral reference backbone (8,458 parameters) and **A5** is the sealed complete configuration (39,500 parameters), which adds a three-route spectral allocation, a simulation-component teacher, multi-task and contrastive objectives, a residual bypass, and a small route-orthogonality regularizer.
- **S**, **M**, and **L** are prospective spectral capacity tiers. S *is* the sealed A5 configuration; M and L widen only the spectral-channel, embedding, and environment dimensions, to 99,596 and 233,244 parameters.
- **sidecar** is a prospective received-I/Q repair branch that raises the total model size from 39,500 to 46,794 parameters.
- **IQFormer-inspired** and **MCLDNN** are architecture-faithful unified-retraining references, not exact reproductions of every original training recipe.
- The **target-support jammer overlap** o is the fraction of the jammer's total STFT power that falls inside the target's 99%-energy time–frequency support. It is not an occupied-bandwidth fraction.

Four evidence levels are kept apart. The original confirmatory family contains the three A5 contrasts. Campaign-internal SNR–SIR, envelope, fusion, and capacity analyses are exploratory. Probe, coverage, route-gating, and sidecar experiments provide subsequent diagnosis or intervention evidence. The severe-held transport test, sidecar repair, inference-time controls, and retrained zeroed-I/Q ablation form four sequential post-campaign stages; each was specified before its own evaluation or retraining,

TABLE I
CELL CONSTRUCTION FOR THE CAMPAIGN-DIAGNOSTIC ENVELOPE. "WINDOWS PER CELL" IS THE RANGE OF SOURCE-DISJOINT TEST WINDOWS CONTRIBUTING TO A CELL.

Split	Role	SIR strata (dB)	Cells	Windows per cell
`id_test`	fit	$-10, -5, 0, +5, +10$	16	230–1053
`hard_interference`	fit	$-15, -10, -5, 0$	16	290–343
`unseen_jammer`	held	$-10, -5, 0, +5, +10$	20	229–276
`unseen_speed`	held	$-10, -5, 0, +5, +10$	20	228–277
`heldout_channel`	held	$-10, -5, 0, +5, +10$	20	216–279
`combined_ood`	held	$-10, -5, 0, +5, +10$	20	223–270
`clean_retention`	excluded	0	1	5000
Fit total			32	
Primary held total			80	

as detailed in Section XVII. Retraining uses only the original campaign training/validation partitions, while the severe-held set remains evaluation-only.

II. Construction of the Evaluation Cells

The envelope is fitted on aggregate cells rather than on individual windows. A cell is one (split, SIR stratum, overlap quartile) triple that retains at least 150 windows and all ten modulation classes. Overlap quartile edges are computed within each split. Table I gives the resulting construction; the fit total of 32 and the primary held total of 80 are both reconstructible from its rows.

Two features of Table I are worth stating plainly. First, the `id_test` split contributes 16 cells rather than 20 because its lowest overlap quartile is populated almost entirely by jammer-free windows, which carry $o = 0$ and the SIR $= 0$ dB cache sentinel; that bin clears the 150-window floor only in the 0 dB stratum. Second, the single `clean_retention` cell is excluded from the primary envelope because a jammer-free window has no physical finite SIR, which is also why its window count is the full split rather than a quartile of it.

III. Campaign-Diagnostic Envelope Skill by Held Regime

Table II reports the held-regime skill for both strong references, including two diagnostics that are easy to omit and easy to misread. "Const." is the fit-domain constant predictor: the mean gain over the 32 fitting cells, applied unchanged to the held cells, with $R^2_{\text{skill}} = 1 - \text{MSE}_{\text{envelope}}/\text{MSE}_{\text{const}}$. "Oracle" is instead the mean of the held cells themselves (pooled row) or of the regime's own held cells (regime rows), which no predictor could know in advance; it is reported as a hindsight reference, not as evidence that the envelope fails to resolve within-regime structure.

The heterogeneity is real and is not smoothed away. Against IQFormer-inspired, the envelope predicts the gain level well in the unseen-speed and held-out-channel regimes ($R^2_{\text{skill}} = 0.75$ and 0.59) and much less well into the unseen-jammer and combined-OOD regimes (0.20 and 0.34); against MCLDNN the unseen-jammer and combined-OOD regimes are at or below the constant baseline. The regime-wise oracle comparison is likewise heterogeneous: on the unseen-speed and held-out-channel shifts the envelope substantially beats the corresponding regime-specific constant (3.94 vs 6.71 and 4.86 vs 6.95 pp against IQFormer-inspired; 4.66 vs 7.78 and 5.41 vs 8.41 pp against MCLDNN), while on the unseen-jammer and combined-OOD shifts the regime-specific constant is better. The pooled oracle (6.04 and 6.81 pp) is lower than the pooled envelope RMSE because it uses the held-domain mean in hindsight; a pooled constant still bears the between-regime variation and the pooled comparison should

TABLE II
HELD-REGIME ENVELOPE SKILL. RMSE VALUES IN PERCENTAGE POINTS.

Reference	Held regime	Cells	Env.	Const.	R^2_{skill}	Oracle	Sign
IQFormer-insp.	unseen speed	20	3.94	7.87	0.75	6.71	1.000
IQFormer-insp.	held-out channel	20	4.86	7.59	0.59	6.95	1.000
IQFormer-insp.	combined OOD	20	7.50	9.23	0.34	4.24	0.950
IQFormer-insp.	unseen jammer	20	7.74	8.67	0.20	3.83	0.950
IQFormer-insp.	pooled (80)	80	6.23	8.37	0.45	6.04	0.975
MCLDNN	unseen speed	20	4.66	8.88	0.72	7.78	1.000
MCLDNN	held-out channel	20	5.41	9.06	0.64	8.41	1.000
MCLDNN	combined OOD	20	10.23	10.28	0.01	2.71	0.900
MCLDNN	unseen jammer	20	10.34	10.24	-0.02	3.23	0.900
MCLDNN	pooled (80)	80	8.10	9.63	0.29	6.81	0.950

TABLE III
CLEAN-SENTINEL FITTING-CELL SENSITIVITY. THE 31-CELL FIT EXCLUDES THE SINGLE `ID_TEST`/SIR$=0$/LOWEST-OVERLAP AGGREGATE DOMINATED BY JAMMER-FREE SENTINEL WINDOWS; THE HELD EVALUATION IS THE UNCHANGED 80 FINITE-SIR INTERFERED CELLS. RMSE VALUES IN PERCENTAGE POINTS; BE DENOTES THE BREAK-EVEN OVERLAP.

Reference	Fit cells	Held RMSE	Const. RMSE	R^2_{skill}	Overlap coeff.	SIR coeff.	BE o @ -15	BE o @ -10
IQFormer-insp.	32	6.23	8.37	0.45	14.38	-1.036	0.11	0.47
IQFormer-insp.	31	6.42	8.28	0.40	15.90	-1.066	0.14	0.47
MCLDNN	32	8.10	9.63	0.29	19.16	-0.894	0.06	0.29
MCLDNN	31	8.20	9.59	0.27	20.55	-0.922	0.08	0.31

not be read as the envelope lacking within-regime structure. The demonstrated property is therefore regime-dependent transport and within-regime structure, not uniform held-domain superiority.

IV. Clean-Sentinel Fitting-Cell Sensitivity

The primary held envelope excludes jammer-free clean retention because a jammer-free window has no physical finite SIR. The fitting domain, however, retains one aggregate dominated by jammer-free sentinel windows, the `id_test`/SIR$=0$/lowest-overlap cell (mean overlap $\approx 10^{-11}$, 1053 windows, SIR taken from the cache sentinel). To close this definition asymmetry, we exclude exactly that aggregate, refit the two-variable envelope on the remaining 31 cells, and re-score the unchanged 80 finite-SIR interfered held cells.

The excluded aggregate is the `id_test`/SIR$=0$/lowest-overlap cell dominated by jammer-free sentinel windows. Excluding it moves the held RMSE by at most 0.20 percentage points and R^2_{skill} by at most 0.05, leaves both coefficient signs and the break-even ordering (required overlap rising as interference becomes less severe) unchanged, and therefore leaves the envelope conclusions materially unchanged; the 32-cell primary fit is retained.

V. Capacity Control across All Splits

The capacity ladder isolates width from representation: S, M, and L differ only in three width numbers and share the routes, teacher, objective, and training recipe. All values in Tables IV and V are computed on the same matched five-seed subset, and must not be mixed numerically with ten-seed results reported elsewhere.

The pattern is consistent across splits. Width raises the average monotonically, and it helps most where the compact model was already competitive: at SIR $=-15$ dB the widest tier gains

TABLE IV
ABSOLUTE MACRO-F1 ON THE MATCHED FIVE-SEED SUBSET. THESE SPLIT-LEVEL FIVE-SEED MACRO-F1 VALUES SHOULD NOT BE NUMERICALLY DIFFERENCED AGAINST THE TEN-SEED CELL-MEAN GAINS IN THE MANUSCRIPT ENVELOPE TABLE.

Split	S	M	L	IQFormer-insp.	MCLDNN
hard interference	0.436	0.455	0.468	0.496	0.454
hard, SIR $= -15$ dB	0.353	0.384	0.403	0.262	0.255
in distribution	0.563	0.584	0.588	0.696	0.636
unseen jammer	0.456	0.462	0.466	0.612	0.584
unseen speed	0.487	0.507	0.510	0.626	0.582
held-out channel	0.488	0.510	0.515	0.613	0.567
combined OOD	0.450	0.462	0.465	0.621	0.587
clean retention	0.500	0.511	0.512	0.693	0.643

TABLE V
PAIRED CAPACITY CONTRASTS ON THE MATCHED FIVE-SEED SUBSET, IN PERCENTAGE POINTS WITH 95% INTERVALS.

Contrast (hard interference unless noted)	Diff.	Low	High
M minus S	1.94	1.07	2.80
L minus S	3.18	2.44	3.96
L minus IQFormer-insp.	-2.84	-4.30	-1.31
L minus MCLDNN	1.37	0.03	2.72
L minus S, SIR $= -15$ dB	4.96	3.48	6.48
L minus IQFormer-insp., SIR $= -15$ dB	14.07	9.55	18.51
L minus IQFormer-insp., clean retention	-18.14	-19.34	-17.00

4.96 percentage points over S and stays 14.07 percentage points ahead of IQFormer-inspired. It does not close the gap that matters for the boundary. On the marginal hard split L is still 2.84 percentage points behind IQFormer-inspired, and on clean retention it is 18.14 percentage points behind, a deficit essentially untouched by tripling the width from M to L (0.10 percentage points of movement). This is the arithmetic behind the manuscript's statement that capacity is not a substitute for the representation change.

VI. PROSPECTIVE DIAGNOSIS AND REPRESENTATION INTERVENTIONS

Table VI compares three follow-up arms designed to distinguish condition coverage, route selection, and representation information. It also provides the intervals omitted from the compact manuscript summary.

The ordering carries the argument. Adding jammer-free windows for every class moves clean macro-F1 by only 0.15 percentage points, providing little support for exposure alone as a sufficient explanation. Gating the modulation route on predicted interference presence moves nothing either, and the learned gate saturates near the modulated route in both clean and hard conditions (99.994% and 99.991%), so it functions as a negative control rather than as a demonstration that a mask fallback was learned. Only the full received-I/Q sidecar configuration moves both clean and hard conditions with intervals clear of zero. The later frozen-inference controls in Section XV show that the effect depends on received-I/Q input at inference; because the lesions remove more than the repair effect itself, they measure dependence and do not quantify the information contribution of the I/Q branch.

TABLE VI
PROSPECTIVE DIAGNOSIS, REPAIR, AND NEGATIVE CONTROL RELATIVE TO A5, IN PERCENTAGE POINTS WITH 95% INTERVALS.

Intervention	Params	Clean diff. [95% CI]	Hard diff. [95% CI]
Per-class clean coverage	39.5k	0.15 $[-0.31, 0.63]$	-0.30 $[-0.91, 0.27]$
Presence-gated route	39.5k	-0.30 $[-0.66, 0.08]$	-0.18 $[-0.72, 0.35]$
Received-I/Q sidecar	46.8k	11.77 $[11.06, 12.49]$	4.37 $[3.44, 5.31]$

VII. SEED-LEVEL ROBUSTNESS

The sealed intervals treat algorithm seeds as fixed blocks. Treating them instead as a random effect and resampling seeds directly gives $[4.02, 5.19]$ percentage points for the A5–A0 whole-configuration contrast, $[-0.15, 0.42]$ percentage points for the teacher-form contrast, $[-0.15, 1.04]$ percentage points for the route-count contrast, and $[-0.19, 0.62]$ percentage points for the residual-free A7 variant against A5. The qualitative reading is unchanged under either treatment: the whole-configuration effect is resolved, the isolated component effects are not.

VIII. COMPARATOR FIDELITY ON A PUBLIC BENCHMARK

This check exists to establish that the two I/Q comparators are implemented competently, not to validate the simulator-domain envelope; a public benchmark cannot do the latter, because it does not expose the separately tracked target and jammer components the overlap covariate requires. The protocol follows the benchmark result table released with the IQFormer reference implementation: the 11-class RML2016.10a set, the full -20 to 18 dB grid, per-stratum $600/200/200$ splits seeded at 233, three algorithm seeds, no augmentation and no input-length change. The public-benchmark fidelity runs use a separate recipe following the benchmark protocol—a 60-epoch budget for IQFormer-inspired and a 200-epoch budget for MCLDNN—and should not be used to infer convergence sufficiency of the 30-epoch main campaign.

Table VII makes the shape of the agreement explicit. Both implementations reproduce the reference release at the level of the all-SNR range, and IQFormer-inspired matches it closely ($+1.04$ percentage points), while MCLDNN remains 3.50 percentage points lower but within the same broad range. Agreement is not pointwise: individual SNR bins deviate by up to about 8 percentage points, which is why this is reported as range-level implementation fidelity rather than reproduction. No fidelity threshold was declared before these runs.

IX. COMPARATOR CONVERGENCE AUDIT

All models were trained under one shared 30-epoch budget with the same cosine-scheduled learning rate and an 8-epoch early-stopping rule; the reported checkpoint is selected solely by validation modulation cross-entropy. Table VIII audits the sealed training histories (selected epoch and completed epochs over the ten algorithm seeds) to check whether any comparator's comparison is distorted by an immediately binding training cap.

The budget binds asymmetrically. IQFormer-inspired and CSSL run the full 30-epoch budget in all ten seeds (median selected epoch 27 and 29), so their reported performance should be interpreted as potentially budget-limited under this shared 30-epoch protocol and the shared epoch budget may favor faster-converging models; the manuscript states this caveat in its Limitations. MCLDNN early-stops in every seed with best epochs well inside the budget (median 13, range 11–16), so the severe-region comparisons against MCLDNN are not explained by a binding cap. A5 itself also trains to the budget

TABLE VII

PER-SNR ACCURACY ON RML2016.10A OVER THREE SEEDS. BRACKETS ARE THE SEED MINIMUM AND MAXIMUM, NOT CONFIDENCE INTERVALS.

SNR (dB)	MCLDNN mean	[min, max]	IQFormer-insp. mean	[min, max]
-20	12.1%	$[12.0, 12.2]$	14.9%	$[14.5, 15.2]$
-18	13.1%	$[12.3, 13.9]$	18.1%	$[16.8, 19.0]$
-16	14.8%	$[14.0, 15.3]$	22.3%	$[21.7, 23.0]$
-14	17.9%	$[16.8, 18.9]$	27.2%	$[26.6, 28.3]$
-12	22.0%	$[20.4, 23.3]$	33.3%	$[31.5, 34.6]$
-10	26.0%	$[25.0, 26.7]$	37.0%	$[36.5, 37.8]$
-8	32.5%	$[31.1, 33.3]$	41.4%	$[41.1, 41.5]$
-6	44.0%	$[42.8, 45.7]$	49.8%	$[49.5, 50.3]$
-4	58.8%	$[58.3, 59.9]$	63.3%	$[63.0, 63.5]$
-2	74.7%	$[74.3, 75.0]$	77.3%	$[76.1, 77.9]$
0	83.4%	$[82.9, 84.1]$	86.1%	$[85.7, 86.7]$
2	85.1%	$[84.7, 85.6]$	89.4%	$[88.6, 90.5]$
4	85.5%	$[84.7, 87.1]$	91.6%	$[90.6, 93.0]$
6	86.1%	$[85.7, 86.8]$	93.4%	$[93.0, 94.1]$
8	86.4%	$[84.9, 88.6]$	93.3%	$[92.8, 94.2]$
10	85.3%	$[84.0, 87.1]$	93.0%	$[92.0, 94.4]$
12	86.3%	$[85.1, 88.4]$	93.8%	$[93.1, 94.7]$
14	86.0%	$[85.0, 87.9]$	93.4%	$[92.9, 93.9]$
16	85.5%	$[84.5, 87.1]$	92.9%	$[92.1, 93.7]$
18	85.2%	$[84.3, 86.8]$	92.9%	$[92.3, 93.9]$
All SNR	58.55%	$[58.1, 59.4]$	65.23%	$[64.8, 65.8]$
Reference release	62.05%	—	64.19%	—

TABLE VIII

CONVERGENCE AUDIT. "AT EPOCH 30" COUNTS SEEDS WHOSE TRAINING RAN THE FULL SHARED BUDGET; "EARLY-STOPPED" COUNTS SEEDS STOPPED EARLIER BY THE PATIENCE RULE. PANEL (A) IS SEALED; PANEL (B) READS THE POST-CAMPAIGN HISTORIES.

(a) Sealed comparator histories

Model	Median best epoch	Range	Seeds at epoch 30	Early-stopped seeds
MCLDNN	13	11–16	0/10	10/10
IQFormer-insp.	27	24–30	10/10	0/10
CSSL	29	26–30	10/10	0/10
A5	29.5	27–30	10/10	0/10
A0	30	28–30	10/10	0/10

(b) Post-campaign capacity and repair histories

Model	Median best epoch	Range	Seeds at epoch 30	Early-stopped seeds
Capacity M	29	27–30	5/5	0/5
Capacity L	27	24–30	5/5	0/5
I/Q sidecar	30	29–30	10/10	0/10

in all ten seeds, and the direct spectral reference A0 likewise runs to the budget in all ten seeds (median selected epoch 30, range 28–30), so the cap concern is not one-sided against the comparators. The post-campaign M, L, and sidecar runs also complete the full budget in every seed, with median selected epochs 29, 27, and 30, respectively; this records their optimization histories without promoting the diagnostic stages into the sealed family.

TABLE IX
SIDECAR LAYER STRUCTURE AND PARAMETER COUNT. ALL CONV1D LAYERS USE STRIDE 1 AND NO BIAS; DW DENOTES DEPTHWISE (GROUPS = IN CHANNELS) AND PW POINTWISE (1×1) CONVOLUTION. TOTAL MODEL SIZE IS $39{,}500 + 7{,}294 = 46{,}794$ PARAMETERS.

Layer	Type	Kernel	Stride	In→Out ch.	Params
1	Conv1D	9	1	2→12	216
2	GroupNorm (3 groups)	—	—	12	24
3	DW Conv1D	7	1	12→12	84
4	PW Conv1D	1	1	12→24	288
5	GroupNorm (6 groups)	—	—	24	48
6	DW Conv1D	5	1	24→24	120
7	PW Conv1D	1	1	24→24	576
8	GroupNorm (6 groups)	—	—	24	48
9	mean/std/max pool + Linear	—	—	72→24	1,752
10	LayerNorm	—	—	24	48
11	Fusion Linear	—	—	72→48	3,504
12	Fusion LayerNorm	—	—	48	96
13	Classifier Linear	—	—	48→10	490
Total sidecar increment					7,294

X. SIDECAR ARCHITECTURE DETAIL

Table IX lists the received-I/Q sidecar layer by layer so the reported parameter increment is fully reconstructible. All convolution layers are bias-free; SiLU activations (parameter-free) follow each normalization. The branch consumes the received two-channel mixture at the full sample rate, uses one depthwise-separable stack with aggressive time pooling (mean/std/max over the window), and fuses the resulting embedding residually with the spectral embedding before a side classification head. The fused embedding passes through the 10-class linear head of layer 13, and these logits are added residually to the base spectral classifier logits; the side head receives no separate auxiliary supervision, and inference consumes only the received mixture. The parameter counts were verified directly against the sealed checkpoint state dict.

XI. DIAGNOSTIC ENVELOPE COVARIATE AND SELECTOR AUDITS

Two zero-compute audits recompute, on the same 32 fitting cells and the same 80 finite-SIR interfered held cells as the primary envelope, (i) a covariate ladder that isolates the skill of each physical covariate and (ii) a diagnostic selector evaluation. Both are exploratory.

Table X gives the ladder. All models are fitted on the 32 fitting cells by least squares and scored unchanged on the 80 held cells; R^2_{skill} uses the same fit-domain constant baseline as the manuscript. SIR is the stronger single covariate against IQFormer-inspired, whereas SIR and overlap have comparable single-covariate skill against MCLDNN; overlap adds a consistent increment beyond SIR against both references, while SIR adds more beyond overlap against IQFormer-inspired than the reverse. The statistical status of that increment is assessed by the paired bootstrap in Section XII.

Table XI evaluates the envelope sign as a diagnostic rule on the 80 held cells. Each sign selector uses envelope coefficients fitted only on the 32 fitting cells and applied to held cells by covariate value; the oracle is a hindsight per-cell best and is not achievable. Realized score is the mean over cells of the selected model's actual macro-F1; MACs use the sealed complexity table values (41.8 M for A5, 355.6 M for IQFormer-inspired).

Inside the held finite-SIR regimes the large I/Q reference leads in almost every cell, so no selector recovers much against always using it, and the envelope's demonstrated value is boundary

TABLE X

COVARIATE LADDER ON THE SHARED 32 FITTING / 80 HELD CELLS. RMSE IN PERCENTAGE POINTS. INCREMENTS: OVERLAP|SIR REDUCES HELD RMSE BY 0.61 PERCENTAGE POINTS AGAINST BOTH REFERENCES; SIR|OVERLAP BY 1.23 PERCENTAGE POINTS AGAINST IQFORMER-INSPIRED AND 0.58 PERCENTAGE POINTS AGAINST MCLDNN.

Model	IQ RMSE	IQ R^2_{skill}	MC RMSE	MC R^2_{skill}
Constant	8.37	0.00	9.63	0.00
SIR only	6.84	0.33	8.71	0.18
Overlap only	7.46	0.20	8.68	0.19
Overlap + SIR	6.23	0.45	8.10	0.29
Overlap + SIR + SNR	6.15	0.46	8.05	0.30

TABLE XI

SELECTOR EVALUATION ON THE 80 HELD CELLS. REGRET IS AGAINST THE HINDSIGHT ORACLE; EXPECTED MACS AVERAGE THE SELECTED MODEL'S COST OVER CELLS. THESE SELECTOR STATISTICS ARE COMPUTED FROM THE TEN-SEED HELD-CELL AGGREGATES AND SHOULD NOT BE NUMERICALLY DIFFERENCED AGAINST THE MATCHED-FIVE-SEED SPLIT-LEVEL MACRO-F1 VALUES IN TABLE IV.

Selector	Realized F1	Regret (pp)	A5 cells	MACs (M)	MAC saving
Always A5	0.4488	14.19	80/80	41.8	88.2%
Always IQFormer	0.5894	0.14	0/80	355.6	0%
SIR-only sign rule	0.5894	0.14	0/80	355.6	0%
Overlap+SIR sign rule	0.5872	0.36	5/80	336.0	5.5%
Oracle (hindsight)	0.5907	0	3/80	343.8	3.3%

localization—identifying the few severe, high-overlap cells where the compact model is competitive—rather than a large expected compute saving in this campaign. The overlap+SIR rule is the only rule that ever selects the compact model (5 of 80 cells), at a regret of 0.36 percentage points against the hindsight oracle; the SIR-only rule never selects it. Computed directly from the held-cell artifacts, its confusion matrix against the hindsight-optimal indicator is TP$=3$, FP$=2$, FN$=0$, TN$=75$: it identifies all three compact-favoring cells (recall $3/3$) with two additional selections (precision $3/5$), so its held-domain value is boundary localization without missed positive cells, while always using IQFormer-inspired still attains the higher average realized macro-F1. This reflects a regime structure in which the compact model's advantage is concentrated in a small corner of the plane.

XII. LEAVE--15-DB-OUT EXTRAPOLATION AND PAIRED OVERLAP-INCREMENT TESTS

Two further zero-compute audits close the remaining envelope questions. Both reuse the frozen cell table.

The 80 held cells contain no SIR $= -15$ dB cell: that stratum occurs only in the `hard_interference` fitting split. To probe extrapolation toward the most severe level, we refit the two-variable envelope on the 28 fitting cells excluding the four -15-dB cells and predict those four cells out-of-sample (a leave-one-SIR-stratum-out test, not an external validation). Table XII compares the envelope with a 28-cell constant and a 28-cell SIR-only fit.

The outcome is reference-dependent, and Table XII must be read on both skill dimensions. For MCLDNN, the full overlap+SIR surface reduces RMSE relative to the constant baseline (6.84 versus 16.43 percentage points) but predicts the correct sign in only two of four held-out cells, whereas the SIR-only surface preserves all four signs (7.87 percentage points RMSE). For IQFormer-inspired, the full model is no better than SIR-only (12.69 versus 12.15 percentage points RMSE; sign accuracy $1/4$ versus $0/4$). The leave-one-SIR-stratum-out test therefore does not confirm a transportable overlap

TABLE XII

ENVELOPE EXTRAPOLATION AUDITS. PANEL (A): LEAVE--15-DB-OUT TEST; ALL PREDICTORS ARE FITTED ON THE 28 FITTING CELLS EXCLUDING THE FOUR HARD_INTERFERENCE SIR $= -15$ DB CELLS AND SCORED ON THOSE FOUR CELLS ONLY; RMSE AND MAE IN PERCENTAGE POINTS; SIGN ACCURACY IS THE FRACTION OF THE FOUR CELLS WHOSE PREDICTED GAIN SIGN MATCHES THE OBSERVED SIGN. PANEL (B): BLOCK-PAIRED BOOTSTRAP OF THE OVERLAP INCREMENT BEYOND SIR ON THE 80 HELD CELLS; ΔMSE IS THE PAIRED SQUARED-ERROR REDUCTION (SIR-ONLY MINUS OVERLAP+SIR), IN pp^2; INTERVALS ARE 2.5/97.5 PERCENTILES OVER 10,000 BLOCK RESAMPLES.

(a) Leave--15-dB-out test

Reference	Predictor	RMSE	MAE	Sign acc.
IQFormer-insp.	Constant	21.34	21.26	0/4
IQFormer-insp.	SIR only	12.15	12.02	0/4
IQFormer-insp.	Overlap + SIR	12.69	12.01	1/4
MCLDNN	Constant	16.43	15.83	0/4
MCLDNN	SIR only	7.87	6.54	4/4
MCLDNN	Overlap + SIR	6.84	6.52	2/4

(b) Block-paired bootstrap on the 80 held cells

Reference	Mean ΔMSE	95% CI	$P(\Delta\mathrm{MSE} > 0)$	RMSE red. (pp)
IQFormer-insp.	7.97	$[-2.06, 18.18]$	0.94	0.61
MCLDNN	10.18	$[-8.29, 29.84]$	0.85	0.61

increment: the full surface does not preserve the severe-region sign structure that the SIR-only surface retains. The -15-dB break-even overlaps remain conditional extrapolations of the fitted envelope rather than independently held-regime-validated thresholds.

For the incremental value of overlap beyond SIR on the 80 held cells, we use a block-paired bootstrap. The paired statistic is the per-cell squared-error difference $d_i = e^2_{\text{SIR-only},i} - e^2_{\text{overlap+SIR},i}$, so $\bar{d} > 0$ means adding overlap lowers MSE. The 80 held cells form 20 held-regime$\times$SIR blocks of four overlap quartiles each; each draw resamples the 20 blocks with replacement ($B = 10{,}000$, seed 20260820) and keeps all quartiles inside a sampled block (Table XII, panel (b)).

Both references show positive point estimates and a majority of positive draws, but both percentile intervals include zero; overlap therefore provides a consistent point-estimate refinement beyond SIR rather than a separately resolved predictor, and the manuscript reports it as an exploratory refinement.

Finally, applying the 28-cell sign rule to the four excluded -15-dB cells with the same ten-seed cell-level aggregates and MAC cost model selects A5 in only one of the four hindsight-A5-favoring cells (sign accuracy $1/4$; realized cell-mean macro-F1 0.285 versus the 0.361 hindsight oracle on those four cells). The cross-model selector therefore transfers poorly to the excluded severe stratum, which is the selector-side counterpart of the extrapolation caveat above.

XIII. Independent Severe-Held Validation: Transfer Boundary of the Corner Advantage and the Diagnostic Envelope

After the campaign, an independently specified robustness evaluation asked whether the severe-corner advantage and the overlap–SIR envelope survive on data generated outside the frozen campaign. The independent severe-held set fixes SIR at -15 dB and uses two source-disjoint regimes with different operating-regime factors: regime R1 keeps the training channel profiles (TDL-A/C/D) and combines them with held jammer families (pulse and OFDM-like emitters) at a held terminal speed of 180 km/h, while regime R2 combines held channel profiles (TDL-B/E) with the same held jammer families at 250 km/h. Each regime contains 6400 fresh sources disjoint from the campaign source pool

TABLE XIII
SEVERE-HELD SEED-MEAN POOLED MACRO-F1 (%) BY JAMMER FAMILY AND BY REGIME, COMPUTED FROM THE SAME FROZEN PREDICTIONS AS TABLE XIV. THE POOLED COLUMN EQUALLY WEIGHTS THE TWO REGIME-LEVEL MACRO-F1 VALUES AND IS NOT THE ARITHMETIC MEAN OF THE JAMMER-FAMILY COLUMNS. MACRO-F1 IS RECOMPUTED WITHIN EACH AGGREGATION, SO THE PER-FAMILY, PER-REGIME, AND POOLED VALUES SHOULD NOT BE NUMERICALLY DIFFERENCED ACROSS AGGREGATION CONVENTIONS.

Model	Pulse	OFDM-like	R1	R2	Pooled
A5	29.03	4.60	21.13	20.31	20.72
IQFormer-insp.	37.45	4.56	23.82	22.77	23.29
MCLDNN	55.34	4.92	33.98	33.52	33.75
Sidecar	38.22	5.74	25.34	24.83	25.09

(12800 new sources against the campaign's 152,000; the disjointness is asserted programmatically at cache build), evaluated on the eight-point SNR grid from -10 to 18 dB at view 0. The set is stored under `standards/cache_s5_severe_held_v1/` with a manifest recording the cache digest. All campaign checkpoints were evaluated unchanged.

Two aggregation conventions appear in this document, and every reported number is labelled by its convention. *Window-pooled* macro-F1 pools all predictions first and then computes macro-F1; it is used for the transfer gains in Table XIV. The *seed-wise* convention computes macro-F1 per seed and per regime first and then averages, with pooled the equally weighted regime mean; it is used for the repair contrast and levels and for all frozen-inference controls, and it is the manuscript's headline convention because it supports paired seed inference. Because macro-F1 is nonlinear, the two conventions need not agree numerically (-3.12 versus -2.58 percentage points against IQFormer-inspired, -12.45 versus -13.03 percentage points against MCLDNN for the A5 gap on this held set), and they are never mixed within a single contrast.

Table XIV reports the transfer results. The pooled A5 gain is negative against both comparators, the envelope's sign accuracy is near chance (and below it against MCLDNN), and a SIR-only or constant predictor beats the full overlap–SIR envelope on RMSE against both references. The result therefore defines a transfer boundary: neither the severe-corner advantage nor the -15-dB envelope transport reproduces on independently generated severe data.

A zero-compute decomposition of the same frozen predictions splits the held set by jammer family and by regime (Table XIII).

The split localizes the failure and explains the level drifts relative to the campaign's SIR $= -15$ dB levels (Table IV). Under OFDM-like interference every evaluated model sits near the floor, while the pulse family remains learnable; A5's collapse is concentrated in the OFDM-like windows. MCLDNN's held level rises above its campaign level because it reaches 55.34% on the pulse family. The held pulse family yields substantially higher I/Q-domain performance under this evaluation: pulse interference is intermittent and leaves near-clean segments inside the window. Because the held set replaces the campaign's training jammer families with these two held families, OOD-ness is confounded with jammer-family difficulty in this comparison. R1 and R2 differ only in channel profile and speed, and the levels move little between them, so the channel-and-speed axis contributes little additional degradation at this SIR. The transfer verdict therefore bounds the joint transport of the corner across compound regime shift and does not attribute the failure to any single factor.

XIV. SEVERE-HELD REPRESENTATION REPAIR BY THE I/Q SIDECAR

After the transfer check revealed a cross-regime transport boundary on the independently generated severe-held set, a second analysis plan specified the repair hypothesis, primary contrast, and decision

TABLE XIV
INDEPENDENT SEVERE-HELD TRANSFER CHECK. POOLED GAIN IS A5 MINUS THE REFERENCE IN WINDOW-POOLED MACRO-F1 PERCENTAGE POINTS. RMSE ROWS APPLY THE FROZEN COEFFICIENTS ESTIMATED FROM THE ORIGINAL CAMPAIGN FITTING DOMAIN WITHOUT REFITTING TO THE INDEPENDENT SEVERE-HELD DATA; THE CONSTANT BASELINE IS LIKEWISE THE FROZEN CAMPAIGN-DOMAIN CONSTANT.

Reference	Pooled gain	Sign acc.	RMSE full	RMSE SIR-only	RMSE const.
IQFormer-insp.	-3.12	0.578	9.75	9.05	8.11
MCLDNN	-12.45	0.406	22.56	22.29	15.41

rule before any sidecar inference on that same held set. The data stay locked to the same cache; the A5, IQFormer-inspired, and MCLDNN predictions are reused from the transfer check; and the ten sidecar checkpoints pass a provenance audit (46794 parameters each, strict loading of the frozen state dicts, SHA-256 recorded in `docs/S5_R1_CHECKPOINT_AUDIT.md`). The primary contrast is sidecar minus A5 under the seed-wise convention, pooled as the equally weighted mean of the per-seed paired differences, with a seed-level paired bootstrap interval.

The repair is consistent across seeds and strata. Seed-mean pooled macro-F1 rises from 20.72% (A5) to 25.09% (sidecar), a difference of 4.37 percentage points ($[2.96, 5.90]$), with regime contributions 4.22 and 4.52 percentage points; the difference is positive in 10 of 10 seeds and in 8 of eight SNR strata. A window-level hierarchical paired audit reproduces the interval. On the same held set the sidecar also overtakes IQFormer-inspired (23.29%; descriptive gap-recovery ratio 1.70) while recovering only 0.34 of the remaining gap to MCLDNN (33.75%). The absolute levels remain low for every evaluated model at this SIR $= -15$ dB operating point, so the result is interpreted as a consistent relative robustness repair. Per-seed, per-SNR, and per-class tables are stored as CSV in the artifact index below. The campaign hard split and independent severe-held sidecar repair both numerically equal 4.37 percentage points, but they are distinct ten-seed contrasts: their intervals are $[3.44, 5.31]$ and $[2.96, 5.90]$ percentage points, respectively.

XV. SEVERE-HELD INFERENCE-TIME ATTRIBUTION CONTROLS

A third analysis plan specified three inference-time attribution controls on the same locked cache before any control inference (`docs/S6_SEVERE_HELD_INFERENCE_CONTROLS_PREREG.md`). These controls probe attribution on the same locked severe-held benchmark.

Control A forwards the frozen capacity tiers M (99596 parameters) and L (233244 parameters) on both regimes, with A5 and the sidecar restricted to the same matched five seeds (17, 29, 43, 71, 101). Tested spectral width scaling does not reproduce the repair: M and L reach 21.77% and 20.89% seed-mean pooled macro-F1 (paired contrasts against A5 of 1.05 and 0.17 percentage points, both intervals including zero), whereas on the matched subset the sidecar reaches 26.11% against 20.72% for A5.

Control B tests the sidecar's information dependence at inference. B1 replaces the I/Q branch input with zeros of the same shape; B2 applies a fixed random temporal permutation of the 1024 samples within each window, identical across the two channels, which preserves the marginal sample distribution but destroys modulation temporal structure. Both interventions degrade the sidecar substantially, to 12.80% and 16.25%, losses of 12.28 and 8.84 percentage points relative to the full sidecar.

Control C removes the residual side-head contribution classifier(fused embedding) and keeps the co-trained base spectral logits. The sidecar falls to 7.09% (-13.62 percentage points against A5),

TABLE XV

FROZEN-INFERENCE ATTRIBUTION CONTROLS ON THE SEVERE-HELD SET (POST-CAMPAIGN DIAGNOSTIC). SEED-MEAN POOLED MACRO-F1 IN %. PANELS SEPARATE THE MATCHED FIVE-SEED CAPACITY COMPARISON, THE FULL TEN-SEED INTERVENTIONS, AND THE INFERENCE-TIME LESION LOSSES RELATIVE TO THE 4.37-PERCENTAGE POINTS REPAIR EFFECT.

(a) Matched five-seed Control A

Condition	Seed-mean pooled macro-F1	Seeds
A5	20.72%	5
Capacity M	21.77%	5
Capacity L	20.89%	5
Sidecar (full)	26.11%	5

All four entries use exactly the same five algorithm seeds (17, 29, 43, 71, 101); contrasts are paired within this matched subset.

(b) Full ten-seed Controls B and C

Condition	Seed-mean pooled macro-F1	Seeds
Sidecar (full)	25.09%	10
Sidecar, I/Q zeroed (B1)	12.80%	10
Sidecar, I/Q time-shuffled (B2)	16.25%	10
Sidecar, side head removed (C1)	7.09%	10

Panel (b) uses all ten frozen sidecar seeds and changes inference only; these results are information-dependence diagnostics rather than retrained architecture ablations.

(c) Inference-time lesion ratios (ten seeds)

Lesion	Level (%)	Loss (percentage points)	Loss / repair
Full sidecar	25.09	—	—
B1 I/Q zeroed	12.80	12.28	2.81×
B2 I/Q shuffled	16.25	8.84	2.02×
C1 side head removed	7.09	17.99	4.12×

Panel (c) reports the inference-time lesion losses against the full sidecar; losses exceeding the repair effect itself disqualify them as information-contribution measurements (see text).

below A5 itself; the retained logits come from a path never trained to classify alone, so the collapse is expected for a jointly trained network and is read as lesion sensitivity below.

Under the frozen decision rule—(a) the sidecar exceeds the widest capacity tier in pooled seed-mean macro-F1; (b) both information losses are at least 1.0 percentage points; (c) the no-head condition retains at least half of the full repair contrast—conditions (a) and (b) hold and (c) fails (-13.62 percentage points retained).

The lesion magnitudes, however, disqualify the inference-time losses as information-contribution measurements (Table XV, panel (c)).

Each inference-time lesion removes 2.02–4.12 times the magnitude of the repair effect itself, and B1 and C1 fall below the unmodified A5 baseline. Condition (c) is recorded as failed under the frozen rule. In retrospect, this criterion is not informative for a jointly trained path because removing the co-trained head moves the network off its training distribution. We retain this outcome and use the retrained zeroed-I/Q ablation in Section XVI to test the remaining attribution question.

The inference-time lesions therefore show sensitivity to the received-I/Q branch and its co-trained head. The retrained variant, in which the I/Q input is permanently zeroed from the start of training, addresses information attribution directly. Per-regime levels and paired contrasts are stored as CSV in the artifact index below.

TABLE XVI
RETRAINED ATTRIBUTION ABLATION EVALUATED ON THE SEVERE-HELD SET. TRAINING USES ONLY THE ORIGINAL CAMPAIGN TRAINING/VALIDATION PARTITIONS; THE SEVERE-HELD SET IS EVALUATION-ONLY. LEVELS ARE POOLED SEED-MEAN MACRO-F1 OVER FIVE MATCHED SEEDS, AND DIFFERENCES ARE AGAINST A5 WITH PAIRED-BOOTSTRAP 95% CIS.

Condition	Level (%)	Difference vs. A5
A5 (reference)	20.72	—
Sidecar retrained, I/Q permanently zeroed	20.18	-0.54 percentage points ($[-1.34, 0.26]$)
Full sidecar (matched seeds)	**26.11**	5.39 percentage points ($[3.54, 7.66]$)

XVI. RETRAINED ZEROED-I/Q ATTRIBUTION ABLATION

The follow-up to the inference-time controls was specified before retraining (Table XVII, `docs/S7_IQZERO_SIDECAR_RETRAIN_PREREG.md`). The zeroed variant keeps the sidecar architecture and its exact parameter count and feeds zeros into the received-I/Q branch at every training and inference step; the spectral path, fusion layer, and side head are trained exactly as in the full sidecar, using the original campaign training and validation partitions, the same recipe, and five matched seeds. The severe-held cache is used only for evaluation after training and never for tuning or early stopping.

On the severe-held set the zeroed variant does not reproduce the repair (Table XVI): its pooled seed-mean macro-F1 is 20.18% against 20.72% for A5, a difference of -0.54 percentage points ($[-1.34, 0.26]$), while the matched full sidecar reaches 26.11%, a gain of 5.39 percentage points ($[3.54, 7.66]$). The prespecified attribution criterion requires the 95% CI upper bound of the zeroed-versus-A5 contrast to lie below $+1.0$ percentage points. The observed upper bound (0.26 percentage points) satisfies this criterion. On the campaign hard-interference split, evaluated on the same five matched seeds, the zeroed variant lands at the A5 level (43.38% versus 43.59%, -0.21 percentage points), while the full sidecar reaches 48.35% on that same five-seed subset. The campaign hard-split result argues against failed optimization. These matched-five-seed levels should not be numerically compared with the ten-seed campaign aggregates reported elsewhere. Within the tested architecture and simulation domain, the gain is attributed to informative received-I/Q input rather than to the added capacity or fusion structure alone.

XVII. PROSPECTIVE-FREEZE PROVENANCE

Table XVII lists the four frozen analysis-plan documents with their SHA-256 digests. The transfer check was frozen before any inference on the new held set; the repair plan was frozen after the transfer result and before any sidecar inference; the control plan was frozen after the repair result and before any control inference; the retrained-ablation plan was frozen after the control outcome and before any retraining. Each document records its primary endpoint and pass/fail rule; the execution logs are kept alongside the operation logs under `docs/`. No commit-level timestamp service was used; the freeze ordering is established by these documents and the operation logs rather than asserted beyond that evidence. The SHA-256 digests certify content integrity, i.e. that each plan document has not been altered after it was written; they do not prove that a document was written before its inference, and that temporal ordering rests on the operation logs.

XVIII. ARTIFACT INDEX

Every number in this document is derived from the frozen artifact tree rather than transcribed from an earlier draft. The relevant entry points are:

TABLE XVII
PROSPECTIVELY FROZEN ANALYSIS PLANS AND SHA-256 DIGESTS. FREEZE ORDER: TRANSFER CHECK, THEN REPAIR, THEN CONTROLS, THEN THE RETRAINED ABLATION. EACH PLAN WAS FROZEN BEFORE ITS OWN INFERENCE OR, FOR THE FINAL STAGE, BEFORE RETRAINING ON THE ORIGINAL CAMPAIGN TRAINING/VALIDATION PARTITIONS; THE SEVERE-HELD SET REMAINED EVALUATION-ONLY.

Stage	Document	SHA-256 (first 16 hex)
Transfer	docs/S5_A_SEVERE_HELD_PREREG.md	2759558b50a046ea
Repair	docs/S5_R1_PREREG.md	058cc76dd693b715
Controls	docs/S6_SEVERE_HELD_INFERENCE_CONTROLS_PREREG.md	be0dc26ccff5b69f
Retrained ablation	docs/S7_IQZERO_SIDECAR_RETRAIN_PREREG.md	d4c9ecccb652231a

- Cell construction and envelope fits: `analysis_zero_compute/outputs/csv/a14_envelope_cells.csv` and `a14_v41_envelope_interfered_holdout.csv`. The V4.4 clean-sentinel 31-cell sensitivity is recomputed from the same cell table by `analysis_zero_compute/a16_clean_sentinel_sensitivity.py`, writing `a16_clean_sentinel_sensitivity.csv` and `a16_clean_sentinel_sensitivity.json`.
- Covariate ladder, selector evaluation, and convergence audit: `analysis_zero_compute/a17_final_s2_audits.py`, writing `a17_covariate_ladder.csv`, `a17_selector_evaluation.csv`, and `a17_convergence_audit.csv` from the same frozen cell table and the sealed per-seed `result.json` training histories.
- Leave-−15-dB-out test, paired overlap-increment bootstrap, and selector confusion matrix: `analysis_zero_compute/a18_final_s4_audits.py`, writing `a18_leave15_out.csv`, `a18_overlap_increment_bootstrap.csv`, `a18_selector_confusion.csv`, and `a18_leave15out_selector.csv` from the same frozen cell table and the sealed per-seed predictions.
- Sidecar architecture: `analysis_zero_compute/tier2_gpu/run_tier2_experiment.py` (`LightweightIQSidecarVIMD`); the parameter counts in Table IX are verified against the sealed checkpoint state dict under `artifacts/tier2_iq_sidecar_v1/models/`.
- Capacity ladder: `tier2_capacity_L_v1_vs_S.csv`, `tier2_capacity_L_v1_vs_M.csv`, `tier2_capacity_L_v1_vs_iqformer.csv`, `tier2_capacity_L_v1_vs_mcldnn.csv`, and `tier2_capacity_M_v2_vs_a5.csv`, all five-seed matched. The common-counter MAC audit is `analysis_zero_compute/v55_capacity_compute_audit.py`, writing `analysis_zero_compute/outputs/v55_capacity_compute_audit.json`.
- Tier-2 interventions: `tier2_iq_sidecar_v1_vs_a5.csv`, `tier2_condition_coverage_v1_vs_a0.csv`, and `tier2_presence_gated_v1_vs_a5.csv`.
- Comparator fidelity: `artifacts/rml2016_10a_fidelity_v3/`, whose `run.json` records the dataset SHA-256, the split protocol, the reference-release values, and the environment.
- The overlap covariate is computed at cache-build time, stored as `overlap.npy` per split, and SHA-256 checksummed in the sealed cache manifest; it is never an input feature of any model.
- Independent severe held cache: `standards/cache_s5_severe_held_v1/`; the manifest records the cache digest and the programmatic source-disjointness assertion against the campaign source pool.
- Severe-held transfer check (Table XIV): `analysis_zero_compute/a19_s5_severe_held.py`, writing `analysis_zero_compute/outputs/a19_s5_severe_held.json` together with the per-cell CSV files, from the frozen predictions under `artifacts/s5_severe_held_v1/`.
- Severe-held repair check: `analysis_zero_compute/a20_s5_r1_sidecar_severe_held.py`, writing the sidecar predictions under `artifacts/s5_r1_sidecar_severe_held_v1/` and `a20_s5_r1_model_levels.csv`, `a20_s5_r1_paired_contrasts.csv`, `a20_s5_r1_per_snr.csv`, `a20_s5_r1_per_class.csv`, `a20_s5_r1_gap_recovery.csv`, and `a20_s5_r1_summary.json`; the checkpoint provenance audit is `docs/S5_R1_CHECKPOINT_AUDIT.md`.
- Severe-held frozen-inference attribution controls (Section XV): `analysis_zero_compute/a21_s6_`

severe_held_controls.py, writing the control predictions under artifacts/s6_severe_held_controls_v1/ and a21_control_levels.csv, a21_control_matched5_levels.csv, a21_control_paired_contrasts.csv, and a21_s6_controls_summary.json; the frozen analysis plan is docs/S6_SEVERE_HELD_INFERENCE_CONTROLS_PREREG.md.

- Severe-held per-family and per-regime decomposition and lesion loss/repair ratios (Table XIII, Table XV): zero-compute analysis_zero_compute/a22_severe_held_decomposition.py, writing a22_severe_held_decomposition.json, a22_per_jammer_levels.csv, a22_regime_drift.csv, and a22_lesion_ratios.csv from the same frozen predictions and the cache manifest's jammer-family records.
- Prospectively frozen retrained zeroed-I/Q ablation (Section XVI): training launcher analysis_zero_compute/tier2_gpu/run_tier2_iqzero_experiment.py, with checkpoints under artifacts/tier2_iq_sidecar_iqzero_v1/; severe-held inference and analysis analysis_zero_compute/a23_s7_iqzero_severe_held.py, writing predictions under artifacts/s7_iqzero_severe_held_v1/ and a23_s7_iqzero_summary.json, a23_s7_model_levels.csv, and a23_s7_paired_contrasts.csv; campaign-level training sanity check analysis_zero_compute/a24_s7_campaign_levels.py, writing a24_s7_campaign_levels.json; the frozen plan is docs/S7_IQZERO_SIDECAR_RETRAIN_PREREG.md.

The reproduction package released with the paper contains the resolved macros, the figure-source CSV files, per-seed aggregates, run configurations, and the full provenance manifest.